



Key factors of industry 5.0-based organizational sustainability

Tímea Czvetkó^{ID*}, Viktor Sebestyén^{ID}, János Abonyi^{ID}

HUN-REN-PE Complex Systems Monitoring Research Group, University of Pannonia, Egyetem str. 10, Veszprém, 8200, Hungary

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ABSTRACT

Organizational sustainability, social responsibility, and governance practices are increasingly important in achieving economic success, long-term sustainability, and overall social well-being. The understanding and transparent communication of organizational sustainability performance is a must to manage risks and identify improvement opportunities. Improvement in such aspects can be achieved through the application of the Industry 5.0 concept and its tools and methods which combine digital tools for supporting sustainable, human-centric, and resilient organizational development. By measuring organizational sustainability performance, target action can be identified in terms of prioritizing Industry 5.0 solutions that contribute to better performance. Therefore, this study provides a network and quality function deployment method-based framework that is eligible to prioritize and analyze which Industry 5.0 solution should an organization focus on to perform better on sustainability requirements. The sustainability requirements are defined based on the interlinked Sustainable Development Goals (SDG) and Global Reporting Initiatives (GRI) indicators that provide transparent organizational communication and benchmarking. The QFD framework allows us to identify the cross-correlation and effects between the requirements and Industry 5.0 solutions and priorities, while its network-based analyses reveal further structural interlinkages to support the identification of the key factors of Industry 5.0 solutions. The applicability of the method is demonstrated through a key relationship exploration study.

1. Introduction

Organizations have a crucial role to play in advancing sustainable development, and measuring and reporting on their impact is key to ensuring progress toward achieving Sustainable Development Goals (SDGs) (Initiatives, 2019). However, organizations often struggle to determine what actions to take and how to implement effective strategies for sustainable development. Many lack the systematic guidance, tools, and methods needed to address complex sustainability challenges and integrate these into their operational and strategic frameworks. Without a structured approach, organizations risk inefficient resource allocation and missed opportunities for impactful change. This paper aims to address this gap by providing a comprehensive tool that offers a standardized framework, systematic methods, and a knowledge repository to support organizations in advancing sustainability efforts and aligning with the emerging Industry 5.0 (I5.0) principles. Industry 5.0 promotes resilient, human-centric, and sustainable operations, organizations must prioritize sustainable development and integrate them into their business models, products, and services (Commission, 2021b). To achieve sustainable development in Industry 5.0,

organizations will need to embrace circular economy principles, reduce waste and emissions, promote renewable energy sources, ensure ethical, equitable, and resilient business practices, and focus on human-in-the-loop (Commission, 2021b). This requires a shift in organizational mindset and culture, where sustainability considerations are integrated into decision-making processes and become a core part of organizational strategy, where corporate social responsibility plays a vital role to support these activities (Rodek et al., 2020). Incorporating Environmental, Social, and Governance (ESG) principles is essential for this transformation, as these principles provide a comprehensive framework that aligns business practices with broader societal goals. ESG principles emphasize environmental responsibility, social equity, and robust governance structures, which are critical for ensuring that Industry 5.0 initiatives are sustainable and inclusive (Asif et al., 2023). Digital technologies can support the transformation of manufacturing systems toward greater circularity and sustainability by enhancing data collection, analysis, and reporting capabilities, enabling organizations to make more informed and ethical decisions (Denu et al., 2023). In this context, decision-making processes should rely on evidence-based

* Corresponding author.

E-mail addresses: czvetko.timea@mk.uni-pannon.hu (T. Czvetkó), sebestyen.viktor@mk.uni-pannon.hu (V. Sebestyén), janos@abonyilab.com (J. Abonyi).

URL: <https://www.abonyilab.com> (J. Abonyi).

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and data-driven methods and techniques to achieve better resource allocation and efficiency (Ghobakhloo et al., 2022).

Therefore, systematically measuring and reporting on sustainability impact is essential, as it will allow organizations to identify their maturity, track their progress and impact of the business decisions on the environment, society, and governance, aligned with sustainable development goals, identify areas for improvement, communicate their sustainability performance to stakeholders, and support benchmarking activities (Rodek et al., 2020). By prioritizing sustainability and embracing Industry 5.0 principles, organizations can contribute to a more sustainable future, and ensure that economic development is achieved in a way that is socially and environmentally responsible. To attain sustainability and conduct performance monitoring, an organization necessitates a *standardized management framework*, a *systematic method*, and a *knowledge repository*.

A standardized management framework provides guidance and structure for mapping progression toward sustainable requirements through Industry 5.0 principles. For this purpose, standard corporate reporting frameworks and the Industry 5.0 concept can be promising. Standard reporting frameworks have been considered such as GRI, SASB (Sustainability Accounting Standards Board), and TCFD (Task Force on Climate-related Financial Disclosures). Based on their general applicability and connection with ESG was one of the main aspect for the selection criteria. SASB is more financial focused and industry specific, and FCFD lacks of other sustainability aspect than climate change. Since GRI has standardized indicators across environmental, social, and governance (ESG) aspects and covers 70% of the indicators (Deloitte, 2023). GRI Standards are providing guidelines for organizations to comprehend and document their effects on the economy, environment, and society in a consistent and trustworthy manner, thereby augmenting the transparency of its contribution toward sustainable development (Initiative, 2021). GRI reporting can be an effective method to assess the sustainable work of companies (Chen et al., 2015) and a significant number of companies use the GRI framework to disclose information on their ESG (Environmental, Social and Governance) activities (Deloitte, 2023). While the Industry 5.0 European initiatives are laying the foundation for transitioning into a more resilient, human-centric, and sustainable organization through the support of integrated digital and technological innovation (Müller, 2020). Organizations should be analyzed through systematic methods integrating these principles and frameworks.

Therefore, there is a need for a systematic method that guides through the process of efficiently defining sustainability requirements and supporting Industry 5.0 solutions, as well as their relationships. For this purpose, the quality function deployment (QFD) method is a promising tool as despite being a quality management standardized method (for Standardization, 2017), quality function deployment enables the identification and quantification of interrelationships and cross-correlations among sustainability requirements and Industry 5.0 solutions, as well as within Industry 5.0. Moreover, comprehending the synergistic effects of the solutions (e.g., through digital transformation tools, enabling technologies, etc.) and how investments in Industry 5.0 solutions facilitate sustainable development and reporting activities within the ESG framework (Tsang et al., 2023) is critical. QFD is said to be an excellent tool for developing organizational processes and supports cultural change and management (Govers, 2001). The QFD method and its extensions have found extensive application in studies related to sustainability, particularly in facilitating decision-making processes. For example in product design (Vinodh et al., 2017), such as for household water treatment system design (Fetanat & Tayebi, 2021), product-service design (Sousa-Zomer & Miguel, 2017), identification of organizational sustainability factors (Hsu et al., 2017) in specific sectors e.g. energy (Haiyun et al., 2021), shipping industry (Lam & Lai, 2015), bioenergy industry (Scott et al., 2015), among others. QFD has been proved to be an efficient tool to analyze complex interactions, such as identifying relationships between sustainability supply chain risks,

supply chain resilience and Industry 5.0 drivers (Hsu et al., 2024). However, these studies reveal the potential in identifying key factors of organizational sustainability design and development, the identified indicators are mostly self-defined and generic, lacking systematic indicators to measure organizational performance. Hence, we propose the adoption of a methodical framework involving the GRI Standard. While there have been analyses of the QFD method and GRI indicators using a performance prism based on concurrent engineering and a developed balanced scorecard, the focus of such analyses has been primarily on economic indicators (Goharshenasan et al., 2022). Thereby, a comprehensive model utilizing multiple aspects of GRI Topic Standards and indicators that measure organizational performance is needed. In addition to assessing organizational performance, a comprehensive overview of tools and methodologies that facilitate the development of organizational sustainability and promote progress toward Industry 5.0 has considerable potential.

An open knowledge repository that includes best practices, tools, and methods for organizations to achieve sustainable requirements has great value and can be further expanded as the field continues to evolve and develop. Open knowledge repositories can provide easy and free access for all, can facilitate knowledge-sharing among organizations, researchers, decision-makers, and other parties, promote innovation, transparency and contribute to the democratization of knowledge (UNESCO, 2021) and support the “leave no one behind” concept of sustainability (Weber, 2017).

Based on the three fundamental aspects, we propose a quality function deployment (QFD) and network analysis-based framework to identify key factors for prioritizing Industry 5.0-related organizational development. The methodology involved a thorough review and restructuring of evaluation systems to establish a structured decision-support framework. QFD has been chosen due to its interpretability and adaptability for dynamic development processes, which have been proved to be efficient. This approach proved operationally viable, particularly for defining priorities within design-focused contexts. Due to the structure of QFD, it ensure that Industry 5.0 developments directly address sustainability challenges by systematically mapping requirements to solutions. However, since the requirements are interlinked, but the QFD cannot reveal those interconnections in a highly extended case. In order to consider the hidden relationships and critical points, network analysis have been utilized. Network analysis can provide higher level information about the dependencies and interconnections between sustainable development goals, requirements and Industry 5.0 solutions.

The sustainability requirements are based on the Global Reporting Initiatives Topic Standards, which are pre-associated with the corresponding SDG targets and reporting requirements (Global Reporting Initiatives, 2022), thereby enabling the assessment of the solutions' effect on sustainable development. To allocate the sustainability requirements and indicators to Industry 5.0 solutions, we promote the utilization of the QFD method. By doing so we are able to identify which sustainability requirements can be supported and developed via which Industry 5.0 solutions, as well as enable us to quantify which solutions are key factors in improving organizational performance on sustainability aspects. The relationships that have been identified are founded upon a thorough analysis of existing literature, documented best practices, readiness models, and European initiatives promoting the progression toward Industry 5.0.

The proposed framework is built upon 17 SDG goals, 61 SDG targets, 141 sustainability requirements (GRI Topic Standards), 18 Industry 5.0 solutions. The applicability of the method is demonstrated through an illustrative study for key relationship exploration that is a partial exploration since each organization must customize the method based on their portfolio. It should be noted that the examples under consideration are not fully exhaustive, as the requirements and solutions are continually expanding through the incorporation of novel aspects and

methodologies, as well as the customization of the variables can include organization-specific strategic and key performance indicators.

Therefore, the contribution and the structure of the manuscript are as follows:

- In Section 3, the method of the QFD and network analysis-based identification of the key factors of Industry 5.0 solutions is discussed.
- Section 4 introduces the applicability and utilization potential of the proposed framework through a partial exploration study to identify key relationships. This section discusses each part of the QFD-based assessment in detail.

We believe that the proposed methodology development holds significant potential for organizations to identify opportunities related to sustainability within the context of Industry 5.0. Furthermore, it can aid decision-making processes related to investment in Industry 5.0 solutions and their impact on organizational performance as measured by the SDGs. By providing a structured approach to mapping key relationships between sustainability requirements and technological solutions, the methodology empowers organizations to pinpoint critical areas for improvement, align their strategies with global sustainability goals, and enhance operational efficiency. Additionally, the framework's adaptability ensures that it remains relevant across diverse industries and organizational contexts, enabling tailored applications that reflect specific goals, challenges, and priorities. As a result, organizations can use the methodology not only to optimize their sustainability strategies but also to strengthen their competitive advantage and long-term resilience in the face of evolving industry demands and global sustainability requirements.

2. Literature review and research relevance

This section provides a general overview of the background of progress toward the integration of aspects of Industry 5.0 into organizational practices and the European Commission initiative to integrate human-centricity and sustainability into corporate reporting (Strauka et al., 2024).

In order to underline the methodological development, a systematic literature review is conducted for the following themes and related targeted queries were considered using the Scopus Database:

1. Progress toward sustainable organizational Industry 5.0 integration:
“Industry 5.0” AND “sustainable development” AND “organization” OR “corporate”
2. Industry 5.0 aspects appearance in corporate sustainability reporting:
“corporate reporting” OR “organizational reporting” AND “sustainability” AND “human”
3. Industry 5.0 tools and methods fostering sustainable organizational development:
“Industry 5.0” AND “organization” AND “technology” AND “sustainable”
4. Utilization of quality function deployment method within Industry 4.0 and 5.0 environment:
“quality function deployment” AND “Industry 5.0”

The Industry 5.0 concept was proposed by the European Commission in 2022 as a European strategy to promote industrial transformation toward sustainability, human-centricity and resilience (Commission, 2021b). This approach also contributes to the European Commission's priorities such as the “European Green Deal” (Fetting, 2020) and the “Europe fit for the digital age” (Komisija, 2021). This joint aspiration steers industries and organizations to adopt human-centric management for digital technologies (DOWN & ACT, 2021), up-skilling and re-skilling workforce (Commission, 2020), pursue resource-efficiency

and sustainable operations (Fetting, 2020), as well as promote continuous innovation and research (Commission, 2021c). The success of transformation relies on the way “societal, organizational and institutional changes are shaped in combination with the technological changes, while taking into account environmental, geopolitical and economic constraints” (Commission, 2024). Since 2022, more than 2700 scientific documents have been published according to the Scopus database in diverse areas with great prominence in computer science, and engineering, followed by business, management and accounting, mathematics, social sciences, decision sciences, economics, econometrics and finance, energy, environmental science, physics and astronomy and others. The diversity and popularity of the Industry 5.0 concept underline the need to continue research in the field.

Recent studies have proven that the successful implementation of Industry 5.0 approaches requires a multi-stakeholder approach, considering not only technological advancements but also social and environmental dimensions to reach sustainable development (Ghobakhloo et al., 2023). This has led to the reshaping of industrial architecture to satisfy requirements of human-centric, collaborative (Tóth et al., 2023), and sustainable features. Empirical findings underlines that the human-centered approaches and developments can lead to improved organizational sustainability (Goswami et al., 2024). According to organizational Industry 5.0, sustainability can drive change in each organizational levels, starting from the actual e.g. manufacturing processes — field level, sensing and control levels, operations management, business planning and supply chain management levels. It can include practices to create work environments that prioritize employee well-being, ergonomics, collaboration with artificial intelligence for more efficient performance (Ruppert et al., 2018). Prioritizing resource efficiency and e.g. optimize energy use (Nauman et al., 2023), reduce waste (Mesjasz-Lech, 2023), implement renewable energy sources (Sreelatha et al., 2024). Beyond the material investments, data-driven and machine learning-based tools and their integration into organizational processes play a critical role in organizational sustainability. For example applying AI in process automation, analysis, optimization or upskilling workforce (Maddikunta et al., 2022). According to Gartner it should be priority to “monitor, prioritize, apply new technologies and improve on existing technologies to enable sustainable business in enterprise outcomes” (Gartner, 2020). However, the integration of these technologies must rely on stable organizational culture and leadership (Saksena & Jha, 2024).

Since sustainability, resilience and human-centricity are not only a pillar of Industry 5.0, but also a strategic aim to integrate sustainability thinking, technology developments, and infrastructure into organizational processes, as well as recommended actions to integrate these approaches into corporate reporting and communication (Strauka et al., 2024). The ISO 26000:2010 Social responsibility also highlights the need for transparent and clear communication about the above mentioned aspects and to maximize organizations contribution to sustainable development (Pojasek, 2011). Even though in the realm of Industry 5.0 sustainable, human-centric and resilient practices are used to gain more competitive and efficient organizational status, it is not only organizational intention, but also a legislative requirement. The European Commission's Corporate Sustainability Reporting Directive (CSRD) regulation requires reports on the social and environmental risks that companies face with, and on how their activities impact people and the environment (Commission, 2025). This aligns with the broader Environmental, Social, and Governance (ESG) framework, which emphasizes corporate accountability in sustainability practices (Commission, 2021a). According to a 2022 survey highlighted that organizations prioritize achieving ESG goals and also found that “90% of executive leaders agreed that digital technology is essential to improving sustainability” (Gartner, 2025). Many companies use the GRI framework to disclose information on their ESG activities, since it is not only cover 70% of the ESG requirements, but also serves as a foundation of the CSRD-based reporting (Deloitte, 2023). It has been shown that

Industry 4.0 technologies, including advanced robotics, additive manufacturing, cloud manufacturing, augmented reality, simulations, the Internet of Things (IoT), horizontal and vertical integration, big data analytics, and cybersecurity, have a substantial impact on GRI standards. Specifically, 85% of environmental, 65% of economic, and 50% of societal GRI standards are influenced by these technologies (Narula et al., 2021). So that digital technologies can drive both economic and sustainable outcomes (Gartner, 2025).

Industry 5.0-related enabling technologies have been linked to sustainability reporting. A system has been presented that supports data gathering and intelligent evaluation of report metrics (Yadav et al., 2024). It includes the application of blockchain, digital twin, big data, machine learning and AI, and IoT technologies. Furthermore, synergies between Industry 5.0 technological capabilities and sustainable performance have been identified and revealed that focusing on technological capabilities boost circular economy practices and sustainable performance supporting the achievement of sustainable development goals (Rejeb et al., 2025). Beyond, for example in manufacturing, a strategic roadmap has been made that reveals the dependencies on Industry 5.0 developments and provide a guidance on promoting sustainable manufacturing (Ghobakhloo et al., 2023).

However, it remains unclear which Industry 5.0-related development supports which specific organizational sustainability requirement, as well as there is no prioritization of which developments should be invested in to meet specific sustainability requirements.

It has been emphasized that a supportive frameworks that facilitate the adoption of Industry 5.0 technologies is required (Goswami et al., 2024). Therefore, our objective is to fill in that gap in research to provide guidance on the impact of Industry 5.0 developments on sustainability requirements and support sustainability reporting practices.

For this purpose, we propose a QFD and network analysis-based framework to determine key factors for prioritizing Industry 5.0-related organizational development. The QFD method demonstrates continued relevance and adaptability within the Industry 4.0 and emerging Industry 5.0 frameworks (Wolniak & Grebski, 2023). It serves as a structured method for translating customer needs and strategic requirements into technical specifications, proving valuable for designing products compatible with these advanced industrial environments (Kar & Rai, 2024) and for selecting or evaluating key enabling factors and technologies (Büyükoğkan et al., 2021) or implementation options (J. et al., 2024). Furthermore, the integration of QFD with digital advancements, such as Digital Twins (Sarka & Valilai, 2023), enhances its application by allowing for virtual simulation and testing of different design parameters and scenarios early in the development process, bridging the gap between customer expectations and the capabilities of digitally interconnected systems.

The application of QFD extends to complex operational areas characteristic of Industry 4.0/5.0, particularly in supply chain management, where it helps mitigate sustainability challenges (Al Amin & Baldacci, 2024) and improve risk resilience (Kar & Rai, 2024), often integrated with multicriteria decision making (MCDM) methods such as AHP (Al Amin & Baldacci, 2024), or FDM-FDISM-ANP-TOPSIS (Hsu et al., 2024). Significantly, in accordance with the emphasis of Industry 5.0 on human centricity and social sustainability, QFD is being used to systematically incorporate social needs and human factors into the design requirements of Cyber-Physical Production Systems (CPPS) (Kumar et al., 2024), ensuring that technological advancements serve broader societal and worker well-being goals alongside efficiency and productivity. Furthermore, QFD has been proved to be an efficient tool to analyze complex interactions between sustainable supply chain and resilience and Industry 5.0 drivers (Hsu et al., 2024), however this application only focuses on supply chain and does not include concrete organizational indicators, and does not linked to reporting practices to measurable performance. However, it is proved that the structure of QFD is eligible to analyze systematic relations between

sustainability requirements and Industry 5.0 solutions, as well as relations between the solutions. Due to the extension of the interlinkages, network analysis methods and metrics have been utilized to gain higher level information. In this way the interdependences and communities can be seen, which indicates the closely connected items. It helps to identify which sustainability requirements (GRI standards) bridge sustainability targets (SDG targets) and Industry 5.0 solutions. Furthermore, it highlights that which Industry 5.0 solutions are the most relevant to specific sustainability goals, and provides guidance in selecting the most suitable solutions. The methodological aspects and steps are detailed in the following section.

3. Network analysis-based method for the indication of key factors of industry 5.0 solutions

By following the proposed systematic development cycle, organizations can ensure that their Industry 5.0 efforts are aligned with sustainable development goals and that they are tracking and reporting on their progress toward these goals and driving long-term sustainability performance.

The sustainability-oriented steps of organizational Industry 5.0 strategy development and reporting are shown in Fig. 1. The concept is partly based on the GRI concept (Initiatives, 2019), which outlines a three-step process to integrate the SDGs in existing business and reporting processes. The concept has been explicated and extended in the context of Industry 5.0.

During the target evaluation stage, it is essential that organizations gain a comprehensive understanding of their environmental requirements, ongoing initiatives, and current developmental stage to establish a baseline for their developmental project. In this regard, identifying the key factors of organizational performance in Industry 5.0, including sustainability, human-centricity, and resilience, as well as the current stage for digital technologies and data-driven solutions, facilitates identifying the areas in which an organization needs to evolve and focus its strategy. Furthermore, with respect to sustainability, comprehending the SDGs and their corresponding targets and indicators in the context of the organization's social, economic, and environmental benefits is crucial for informed decision-making. Based on this information, organizations can prioritize the targets and areas in which they aim to make an impact and potentially invest in. The prioritization of efficient performance toward achieving SDGs by implementing digital technologies and Industry 5.0 approaches aims to enhance productivity and economic benefits through the reduction of waste and environmental impact in industrial processes, and the implementation of cleaner production practices and sustainable manufacturing (Ching et al., 2022). Such sustainable practices can lead to financial benefits, such as cost savings through optimized processes, reduced waste, and energy, and increased efficiency (Piyathanavong et al., 2019). Additionally, it can lead to enhanced reputation among customers and investors, thereby fostering brand loyalty (Quintana-García et al., 2022).

In the measure and analyze stage, organizations are required to establish their business objectives, select appropriate performance indicators, and undertake a comprehensive analysis of their performance. The chosen indicators must be evaluated within the context of Industry 5.0 while developing a strategic developmental plan. In this regard, our proposed methodology provides an excellent base pairing sustainability indicators with Industry 5.0 solutions and calculates which solutions a given organization should focus on/invest in based on their set preferences and additional capacity constraints.

In order to support Industry 5.0 strategy-making, our proposed method can be integrated into the model development practices to identify the relationship between sustainability requirements and Industry 5.0 solutions. For this purpose, the foundation of the quality function deployment method is utilized.

Fig. 2 indicates the schematic representation of the quality function deployment-based structure and assessment of supporting the development of organizational sustainability.

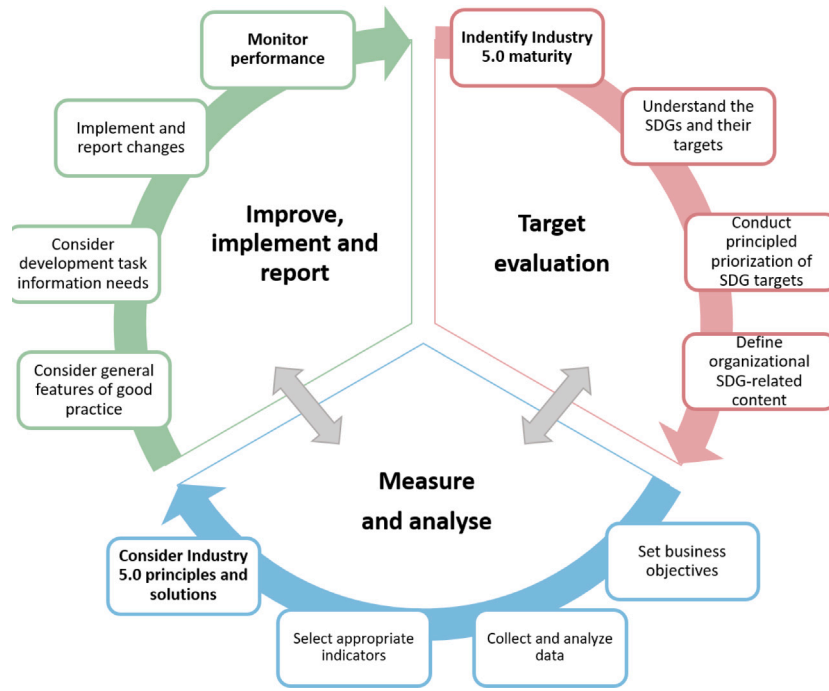


Fig. 1. Sustainability-oriented steps of organizational Industry 5.0 strategy development and reporting based on Ref. Initiatives (2019).

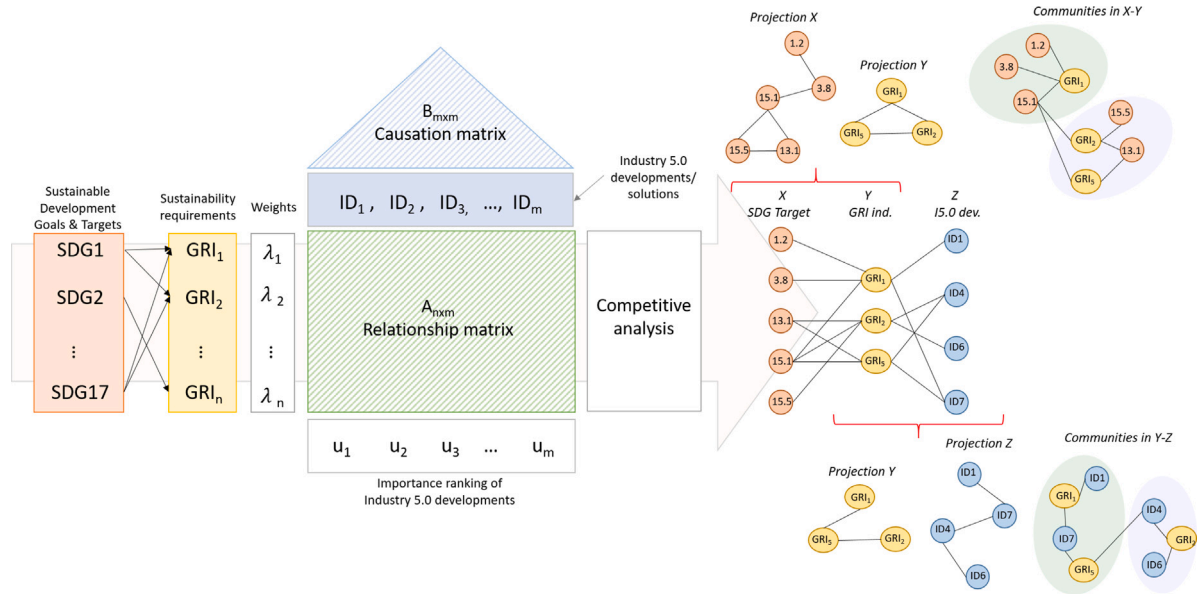


Fig. 2. Schematic structural representation of quality function deployment for organizational sustainable development.

The QFD for organizational sustainable development is built as follows:

- The 17 sustainable development goals and the related indicators have been connected to the sustainability requirements based on the Global Reporting Initiatives report on linking the SDGs and the GRI Standards (Global Reporting Initiatives, 2022).
- The sustainability requirements are gathered based on the GRI Topics (Global Reporting Initiatives, 2022). Let GRI_i be a set of sustainability requirements. Each GRI_i is weighted with λ_i , $i = 1, \dots, n$. The selection of weights can be determined according to the organizational preference, specifically, the sustainability requirements that are targeted for emphasis.

- Industry 5.0 solutions are identified by the work (Ghobakhloo et al., 2022), which however covers organizational, system, functional and managerial level solution, but the human-in-the-loop aspects seems to be missing. Therefore, the solutions have been extended with the concept of Operator 4.0 (Ruppert et al., 2018) and Human-centric approach. The former represents the functional and technological field on the shop floor, the latter the managerial and strategic aspects of providing foundations to the developments. The identified 18 solutions are represented by ID_j , where $j = 1, \dots, m$ is the j th solution.
- The relation of the sustainability requirements and I5.0 solutions form an $n \times m$ matrix A . $a_{i,j}$ ($i \in GRI, j \in ID$) = 1 if there is a relationship between the i th sustainability requirement and j th I5.0 solution, 0 otherwise.

- The relationship of the defined I5.0 solutions is explored in an $m \times m$ B matrix, where $b_{j,k}$ highlights how would the change in solution j effect solution k . These interlinkages are based on the work of Ghobakhloo et al. (2022). Note, that $b_{j,k} \neq b_{k,j}$ and $b_{j,j} = 1$. We assume the synergies between I5.0 solutions have a one-way causal effect. If $b_{j,k} = 1$, then b_j tend to directly case b_k . In case $b_{j,k} = 1^*$, where the “*” indicates that b_j indirectly have a causation effect on b_k through other I5.0 solutions.
- The importance ranking value of the I5.0 solutions is calculated as $u_j = \sum_{i \in I_D} \lambda_i a_{i,j} \quad \forall i \in GRI$, where u_j is the importance rank of the j th Industry 5.0 solution, and $j = 1, \dots, m$.
The importance ranking aims to provide an outline of how much Industry 5.0 solutions tend to influence the selected sustainability requirements and which solutions should the organization focus on developing. The greater the value of u_j , the Industry 5.0 solution is more important on the sustainability requirements. If the u_j is high, it either occurs affects important sustainability requirements (with high λ value) or happens to be in a relationship with several sustainability requirements. We must note that the value of u_j does not consider the interplay between the Industry 5.0 solutions and their effect on each other indicated in the matrix B .
- Finally, the features of other organizations are compared based on suggested indicators of the GRI standards (Global Reporting Initiatives, 2022) and extended by experts in the field. It should be noted that these indicators are subject to further refinement and expansion as the field continues to evolve and develop. In this way, organizations can be positioned based on their sustainability performance and identify specific development strategies.
- The proposed QFD can be visualized as a network, enabling structural analysis through the interconnections of SDGs, GRI, and I5.0 developments. By representing these interlinkages as bipartite graphs (Zha et al., 2001), the indirect relationships and interactions can be more effectively illustrated. A bipartite graph is a type of graph in which the set of vertices (or nodes) can be divided into two distinct, non-overlapping groups such that no two vertices within the same group are connected by an edge. Instead, all edges connect vertices from one group to the other. The network is then denoted as $G = (V, E)$, where G is the network, V represents the vertices. V vertex set can be partitioned into non-empty subsets X , Y and Z , $X \cup Y = V$ and $X \cap Y = \emptyset$ and $Y \cup Z = V$ and $Y \cap Z = \emptyset$. Edges are indicated with E , and each edge of G has one endpoint either in X and one endpoint in Y , or one endpoint in Y and one in Z . The partition $V = X \cup Y$ or $V = Y \cup Z$ are the bipartitions of G . In this case, the subsets will be X , the SDG indicators, Y , the GRI indicators and we can denote Z as the Industry 5.0 solutions. By structuring it this way, indirect relationships across all subsets become more apparent, enabling a deeper understanding of interdependencies.
- Communities in networks can be detected, where the connection between the nodes is more dense indicating close interdependence. The Louvain method (Blondel et al., 2008) is used for community detection to extract non-overlapping communities within each bipartite network.
 - SDG Targets - GRI Topic Standards network:
Communities reveal alignment between sustainability goals and targets and organizational reporting initiatives, supporting the identification of where organizational efforts support the SDGs and where gaps exist.
 - GRI Topic Standards - Industry 5.0 solutions network:
Communities highlight synergies between organizational sustainability reporting and development solutions, guiding strategic collaboration and increasing impact.

Understanding these communities can lead to more targeted, strategic actions in sustainability reporting and implementation, benefiting both organizations and broader global development efforts.

The systematic analysis of the interplay between SDGs, GRI Topics/sustainability requirements and the potential Industry 5.0 solutions provides a deeper understanding of the synergies between these elements. However, given the numerous potential relationships between SDGs, requirements, and solutions, a QFD-based analysis may become complex due to its extensive nature. Consequently, a network-based representation and evaluation can offer supplementary insights and reveal bridges between sustainability goals, requirements and Industry 5.0 solutions. By doing so, organizations can better prioritize their development efforts, improve their competitive positioning, and increase transparency, all of which contribute to long-term value creation. Although, it is important to emphasize that this method cannot be fully understood in isolation; rather, it requires experts within the organization to utilize the acquired information to formulate a tailored strategic plan for the organization.

4. Identifying and analyzing the effects of industrial 5.0 solutions on organizational sustainability

In this section we aim to provide a comprehensive overview, which includes general observations and highlights the customizable nature of the proposed method. Furthermore, the example model is not tailored to any single organization; however, its behavior and outcomes are briefly examined in the context of specific organizations in certain cases.

The proposed interactions have been identified for 17 SDGs and 61 SDG targets, 141 GRI Topics, 18 Industry 5.0 solutions. To demonstrate the applicability of the method, an explorative study for identifying key relationships between sustainability requirements and potential Industry 5.0 developments if formed, which is a selected submodel of the whole identified model by including only seven GRI Topics in the study, namely the General Disclosure, Economic Performance, Indirect Economic Impact, Market Presence, Production Practices, Anti-competitive Behavior, Materials, Energy, Waste, and Employment. The Industry 5.0 solutions were identified based on the work of Ghobakhloo et al. (2022), who identified 16 sustainable development functions of Industry 5.0. Those proposed solutions cover areas of Operational and resource efficiency (ORE), Open sustainable innovation (OSI), Renewable integration (RIN), Real-time communication (RTC), Supply chain adaptability (SCA), Supply chain modularity (SCM), System integration and interoperability (SII), Service-orientation and personalization (SOP), Sustainable thinking (STH) Business process monitoring (BPM), Upskilling and reskilling (UPR), Value network integration (VUI), Data sharing and transparency (DST), Intelligent automation (INA), Employee technical assistance (ETA), Circular intelligent products (CIP). However, we must note, that Industry 5.0 transformation requires a balanced “technological advancement with human-centric values, emphasizing the importance of resilience, sustainability, and the well-being of workers” (Commission, 2024). In this way, to give an additional focus on the human-in-the-loop aspect, we have been extended this work with the Operator 4.0 and Human-centric management. Even though the Operator 4.0 concept is rooted in Industry 4.0, it remains highly relevant in the Industry 5.0 context due to its focus on empowering workers through advanced technological support. The Operator 4.0 concept reflects to the functional, technological developments on the shop floor and integrates the aspects of how emerging technologies can assist the work and well-being of operators. This aligns directly with the specific technological inputs detailed in Industry 5.0 Community of Practice (CoP 5.0), such as Computation, AI/ML, Robotics, Wearables, etc. (Commission, 2024) and the technical interaction skills required (Commission, 2024). Furthermore, technologies can be categorized based on the which skills of the operation

we want to empower, e.g. analytical skills, augmented support, collaboration and social interactions, health and well-being. Technologies such as smart glasses, personal displays, smart watches, tablets, smart gloves and clothes, ergonomic and adjustable work environment, or e.g. motion detection provide solutions to ease operators physical and mental workload as well as provides data-driven feedback on performance (Ruppert et al., 2018). Complementing this technological focus, Human-centric management addresses the higher level managerial and strategic aspects of creating sufficient workplace culture that prioritizes inclusivity, collaboration, and continuous learning. This aligns with CoP 5.0 human centricity in terms of including the necessary workforce competencies and mindsets, such as strong entrepreneurial mindset, inter-cultural, inter-disciplinary and diversity-oriented mindset, etc., the establishment of a common vision and inclusive culture, and the crucial organizational processes that foster learning, well-being, participation, and ethical technology deployment (Commission, 2024). It serves as a foundation of prioritizing workers' both physical and mental well-being, empower their skills. This approach also includes leadership development (Thomas et al., 2023), ethical decision making and transparent data management (El-Annan & Hassoun, 2025) practices. By integrating these principles, organizations can create a sustainable and resilient workforce, where technological advancements complement human capabilities rather than replace them, ultimately driving innovation and long-term success. We must note that each solution can be further broken down into technical developments that contributes to perform on the given Industry 5.0 solution. Within the Industry 5.0 framework, technological solutions of Operator 4.0 and organizational human-centric management strategies must intertwine and complement each other to create human-centered, resilient, and sustainable work environments.

The submodel was selected based on expert experience in Hungarian manufacturing organizational collaborations. The QFD-based representation of organizational sustainability development for the study is shown in Fig. 3. Each part of the QFD-based model is presented, and the connections between the elements are discussed in the indicated subsections.

4.1. Connecting SDG targets to sustainability requirements

The study includes ten GRI Topics, from which the following seven are selected by being weighted with one: General Disclosure, Economic Performance, Indirect Economic Impacts, Materials, Energy, Waste and Employment. Note, that the weights (λ) assigned to the topics in the model are not restricted to binary values of 1 and 0. Rather, the weights can be tailored to the specific needs of the organization, allowing for greater flexibility in capturing the relative importance of each topic. Accordingly, a topic deemed to be of greater significance within the organization may be assigned a higher weight to reflect its relative importance.

The selected sustainability requirements are linked to SDG targets based on the work of the United Nations and the Global Reporting Initiatives (Global Reporting Initiatives, 2022) as shown in Table A.1. Furthermore, the indicators of measuring individual organizational performance on the selected requirements are indicated in Table B.1.

In Fig. 4, the summary distribution of the number of GRI Topics and GRI Indicators by SDGs are shown for (a) the whole model, and (b) the smaller-scale study with the selected requirements.

According to the selected requirements, SDG 8 (Promote sustained, inclusive and sustainable economic growth, full and productive employment and decent work for all) and SDG 12 (Ensure sustainable consumption and production patterns) have connections with the most selected GRI Topics thereby they are the most connected SDGs. However, concerning the full scale of GRI Topics (not limited to the application study), the most connected SDGs are SDG 8, SDG 16 (Peace, justice and strong institutions), SDG 3 (Good health and well-being), SDG 6 (Clean water and sanitation), and SDG 12.

The assessment of economic growth (SDG 8.1) can be conducted by examining the economic performance of organizations. This can be achieved by measuring the direct economic value generated by the organization, which is typically represented by organizational revenues. Additionally, economic value distribution must also be taken into account. This includes various components such as operating costs, employee wages, and benefits, payments to providers of capital, payments to the government by country, and community investments (GRI 201-1) (Global Reporting Initiatives, 2016a).

Development-oriented policies on supporting creative, innovative productive activities, job and enterprise creation (SDG 8.3) and equity for all (SDG 8.5) have the potential to foster local economic development by establishing flexible markets that can readily accommodate fluctuating consumer demands while effectively reflecting the costs of goods (Sultana et al., 2022) and the indirect economic impacts (GRI 203-2) (Global Reporting Initiatives, 2016b).

To promote sustainability and economic productivity, it is important to approach the creation of organizational value from a sustainability perspective (Grabowska et al., 2022). This includes forming strategies including diversification, technological upgrading, and innovation (SDG 8.2). Digital advancements such as real-time communication, data sharing, and monitoring can be utilized to optimize organizational processes across the value network, leading to improvements in energy consumption, production efficiency, and environmental performance of products throughout their life cycle. Moreover, there is a growing demand for smart products and circular smart manufacturing (Stock & Seliger, 2016), as well as a system-based approach for organizational circular economy (Halloui et al., 2022).

Circular economy models are also gaining increasing attention due to their potential to mitigate climate-related risks (GRI 201-2) (Global Reporting Initiatives, 2016a), reduce waste generation (GRI 301-1, 301-2, 301-3) (Global Reporting Initiatives, 2016c) and related impacts (GRI 306-1, 306-2, 306-3, 306-4) (Global Reporting Initiatives, 2020), decrease energy consumption within and outside of the organization (GRI 302-1, 302-2) (Global Reporting Initiatives, 2016d), reduce the energy requirements of products and services (GRI 302-4, 302-5), and promote efficiency in sustainability management, consumption, and production while decreasing the material footprint (SDG 8.4, SDG 12.2) (Dantas et al., 2021).

Responsible consumption and production are fundamental pillars of sustainable development as they directly affect the environment. The production processes and consumption patterns have environmental impacts, and hence there is a need to make these processes more responsible. The ultimate goal is to reduce the environmental impacts of production and consumption activities while ensuring equitable access to adequate resources for all (Küfeoğlu, 2022). Therefore, to promote smarter product use and manufacturing, extend the lifespan of products and their parts, and useful application of materials, the following strategies should be considered during strategy-making, such as recover, recycling, repurpose, remanufacture, refurbish, repair, reuse, reduce, rethink, and refuse (Morseletto, 2020). It includes reducing waste generation through prevention, reduction, recycling, and reuse (SDG 12.5) (GRI 301-1, 301-2, 301-3), and minimizing their adverse impacts on human health and the environment (SDG 12.4) (GRI 306-1, 306-2, 306-3, 306-4).

The consideration which SDGs and sustainability requirements (GRI Topic Standards) should be measured within an organization can be also industry specific. For example for automotive industry it has been revealed that the most commonly reported SDG goals (based on seven manufacturers) are the SDG6 - Clean water and sanitation, SDG8 - Decent work and economic growth, SDG9 - Industry, innovation and infrastructure, SDG10 - Reduced inequalities, SDG 11 - Sustainable cities and communities and SDG12 - Responsible production and consumption (Perello-Marin et al., 2022). Also the focus points may differ not just industry, but country and institutional factors such as politics an low, economics and finance, society and culture, technology and

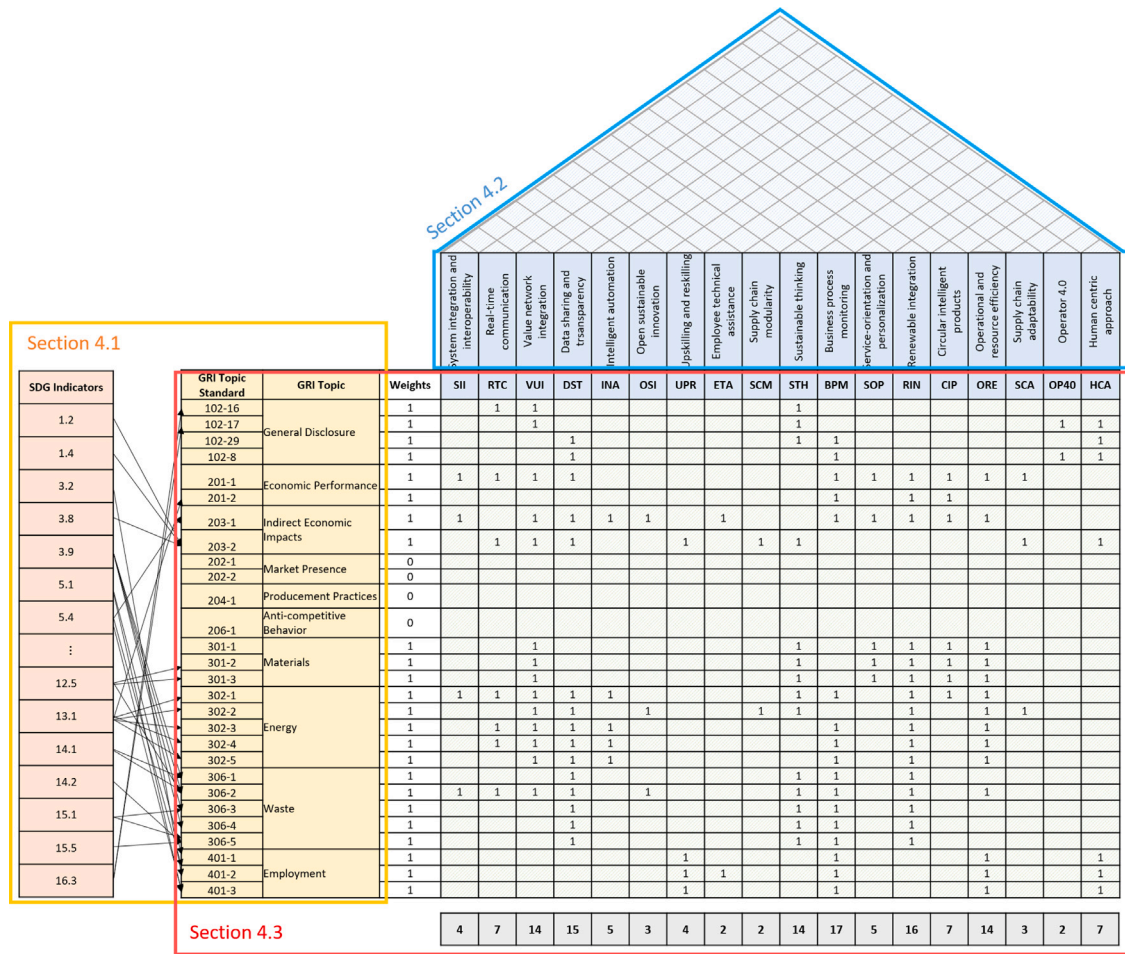


Fig. 3. Framework of QFD-based sustainability-focused key Industry 5.0 solution exploration.

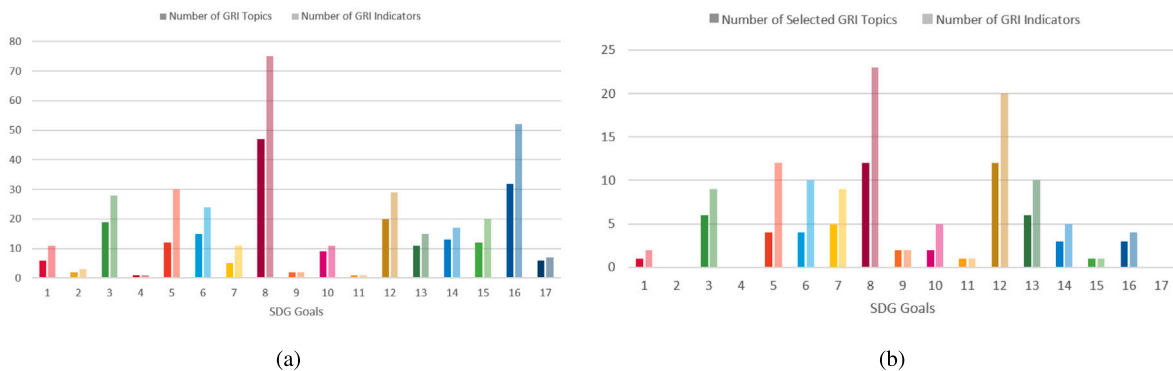


Fig. 4. Number of GRI Topics and GRI Indicators by SDGs for (a) the whole model; (b) the submodel utilized for the study.

innovation, education and labor, and sustainability (Rosati & Faria, 2019).

We must note that the transition toward a sustainable and human-centered economy relies on customer, business, and government parties' perceptions and efforts. A change in consumption patterns is not solely enough, but the environmental impacts of production must be reduced, and regulated through governmental policies and initiatives in investing in digital technologies supporting sustainable development aspirations (Küfeoğlu, 2022). Digital and disruption technologies can contribute to the realization of SDGs and bring a broad range of implications for organizations, customers and the workforce (Kasinathan et al., 2022). The emergence of technologies such as AI-, data-,

model-driven developments, robotics, IoT, etc. poses a challenge to both leaders and innovators, requiring them to reevaluate their organizational and business strategies for the future (Kasinathan et al., 2022).

4.2. Causation matrix of industry 5.0 solutions

The causation matrix of Industry 5.0 solutions identifies their relationship. We assume a one-way causal effect in which information is based on the work of Ghobakhloo et al. (2022) and have been extended by the authors with the Operator 4.0 (Ruppert et al., 2018) and Human-centric management.

Table 1
Causation matrix of Industry 5.0 solutions.

		BPM	CIP	DST	ETA	INA	ORE	OSI	RIN	RTC	SCA	SCM	SII	SOP	STH	UPR	VUI	OP40	HCA	Driving force
Business process monitoring	BPM	1	1	0	0	0	1	0	1*	0	1	0	0	1*	0	0	0	0	0	6
Circular intelligent products	CIP	0	1	0	0	0	1	0	1	0	1	0	0	0	0	0	0	0	0	4
Data sharing and transparency	DST	1	1	1	1	1*	1*	1	1	1*	1	0	0	1*	1	1	1	0	0	14
Employee technical assistance	ETA	1	1*	1*	1	0	1	1*	1*	1*	1*	0	0	1*	1	1*	0	1	0	13
Intelligent automation	INA	1	1*	1	1	1	1	1*	1	0	1	1	0	1*	1*	1	1*	1	0	15
Operational and resource efficiency	ORE	0	0	0	0	0	1	0	0	0	1	0	0	0	0	0	0	0	0	2
Open sustainable innovation	OSI	1*	1	0	1	0	1*	1	1	0	1	0	0	1	1*	1	0	0	0	10
Renewable integration	RIN	0	1	0	0	0	1*	0	1	0	1*	0	0	1*	0	0	0	0	0	5
Real-time communication	RTC	1	1	1	1	1	1*	1*	1	1	1	1*	0	1	1*	1*	1*	0	0	15
Supply chain adaptability	SCA	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	1
Supply chain modularity	SCM	1	1*	0	0	0	1	0	0	0	1	1	0	1	0	0	0	0	0	6
System integration and interoperability	SII	1	1*	1	1	1	1*	1*	1	1	1	1	1	1	1*	1*	1	1	0	17
Service-orientation and personalization	SOP	1*	1	1*	0	0	1	0	1*	1*	1	0	0	1	0	0	0	1	0	9
Sustainable thinking	STH	1*	1	1*	1*	1*	1*	1	1	1*	1*	0	1*	1	1	1	1*	1	1	17
Upskilling and reskilling	UPR	1*	0	1*	1	0	1*	0	0	1*	0	0	0	0	1*	1	0	1	0	8
Value network integration	VUI	1	1	1	1*	1	1	1	1	1	1*	1*	0	1	1*	1*	1	1	0	16
Operator 4.0	OP40	1	1*	1	1*	1*	1	1*	1*	1	0	0	1*	1*	1*	1*	1*	1	0	15
Human-centric approach	HCA	1	1	1	1	1	1	1	1	1	0	0	1	1	1	1	1	1	1	16
Dependence force		14	15	11	11	8	17	10	14	10	15	5	4	14	11	11	8	9	2	

The causation matrix indicates, how would a change in ID_i affect ID_j . This information is stored in a B matrix as shown in Table 1, which is the causation matrix of the application study. Specifically, if $b_{j,k} = 1, j, k = 1, \dots, 18$, solution ID_j has a direct causal impact on solution ID_k , as evidenced by the example of Circular intelligent product (CIP) directly affecting Operational and resource efficiency (ORE), while the reverse is not true. If $b_{j,k} = 1^*$, solution ID_j has an indirect causal influence on ID_k through other Industry 5.0 solutions. For instance, Data sharing, and transparency (DST) have a direct effect on CIP, which in turn directly impacts ORE, thereby establishing an indirect causal connection between DST and ORE.

Industry 5.0 solutions considered in the analysis and their short description and importance regarding Industry 5.0 initiatives are as follows based on the work of Ghobakhloo et al. (2022). System integration and interoperability (SII) involves combining micro-modules of digital ecosystems within and across smart components of Industry 5.0, and enabling seamless communication between them (Sharma et al., 2022). Specific solutions include digital twins, LiDAR for precision mapping, and IoT platforms for interconnected operations. SII has important implications for sustainable development, as it provides stakeholders with open, democratized, access to value, and facilitates Real-time communication (RTC) and Value network integration (VUI). Value network integration involves horizontal and vertical integration of business processes throughout the whole supply chain, which in the

case of Industry 5.0 can refer to the ISA-95 standard-based organizational structure (vertical) (of Automation, 2023), and Industrial Data Spaces allowing horizontal communication and cooperation between supply chain members for reaching a common goal (Association, 2019). Key enablers include blockchain for secure data exchange, AI-powered predictive analytics, and cloud-based ERP systems for supply chain synchronization. Thereby, VUI facilitates the integration of optimization algorithms, product customization, manufacturing agility, and collaboration (Csalódi et al., 2021), which are outcomes essential to the sustainability pillars of Industry 5.0. In this context, Real-time communication plays a vital role as provides seamless communication and data exchange between smart components in the value network to improve production reliability, work environment safety, risk management, and decision processes (van de Sand et al., 2019). Technologies like 5G networks, edge computing for latency reduction, and machine-to-machine communication protocols are essential for RTC. RTC is also a critical component of establishing Data sharing and transparency (DST), which refers to the practice of making data available, and accessible to relevant stakeholders in a transparent and secure manner (Borgogno & Colangelo, 2019). Specific tools include standardized ontologies, secure data lakes, and privacy-preserving analytics techniques. It requires a strong base of integrated standardized data models to ensure transparency and interoperability (IDS, 2024). By doing so, it allows better decision-making, and cross-collaboration for a common goal

such as reducing environmental impact or monitoring environmental performance across the value network (Gartner, 2021). DST and VUI enable to perform Open sustainable innovation (OSI) through close collaboration among the value network. OSI can support the operation of developing new products, services, and business models that contribute to sustainable development (Bigliardi & Filippelli, 2022), and digital green innovation (Yin & Yu, 2022).

However, the developed systems, products and services must be continuously monitored and developed to remain competitive and efficient. To do so, Business process monitoring (BPM) through advanced technologies, real-time communication, data-driven analysis, and machine learning methods enables to track, analyze, and optimize business processes across the value chain and monitor key performance indicators, and supports data-driven decision-making, and early risk identification to improve overall business efficiency and effectiveness (Czvetkó et al., 2022). Examples include process mining tools, KPI dashboards, and anomaly detection systems powered by AI. The adaptation of BPM requires an integrated data and information sharing environment and advanced technologies. The integration of Intelligent automation (INA) involves the implementation of such technologies as artificial intelligence, machine learning, and robotics enable to create an automated, adaptive and sustainable production environment (Jafari et al., 2022). Specific solutions include collaborative robots (cobots), autonomous mobile robots (AMRs), and automated quality inspection systems. In the context of Industry 5.0, it aims to achieve greater efficiency, flexibility, and resilience in manufacturing processes, leading to improved process, worker and resource productivity and reduced social and environmental impact (Jafari et al., 2022). INA is also identified as a direct enabler for Employee technical assistance (ETA) and Upskilling and reskilling (UPR).

ETA refers to the use of intelligent systems and digital technologies to provide technical support and assistance to workers and ensures that they are adequately trained to work with complex (cognitive/collaborative) systems (Xu et al., 2021). Tools include augmented reality (AR) interfaces, dynamic work instructions, wearables, smart glasses and watches, personal displays, and virtual training simulators. Therefore, regarding UPR, the continuous development of skills and knowledge of the workforce is a must to meet the evolving needs of Industry 5.0 (Orso et al., 2022). The Industry 5.0 concept emphasizes the human-centric management (HCA) that prioritizes empowering workers and create collaborative system where utilize technologies to enhance their efficiency, safety and well-being (Commission, 2021b). A manufacturing worker in such environment called Operator 4.0, who is equipped with digital skills and utilize e.g. augmented reality, wearables, and collaborative robots (Romero et al., 2020). Examples include AR-enhanced (e.g. maintenance) tools and robotic exoskeletons to assist in physically demanding tasks, as well as performance monitoring, movement analysis, support of 5S workplace organization and digital twins. The HCA also triggers Sustainable thinking (STH), which refers to the integration of sustainability principles through the value network and takes into account social, economic, and environmental factors. Incorporating these principles, the Renewable integration (RIN) concept utilizing smart and renewable sources and materials for product design, as well as integrating renewable principles into the entire organizational culture are crucial features (Duarte et al., 2020). Solutions include energy harvesting technologies, biodegradable materials, and life cycle assessment tools. In this regard, Circular intelligent products (CIP) involve using Industry 5.0 technologies to design products that can function within the circular economy framework. Technologies like reverse logistics platforms, additive manufacturing for remanufacturing, and AI-based material recovery systems are pivotal. CIP ensures that products are designed and developed to be environmentally friendly across their life cycles (Turner et al., 2022). The overall organizational objective utilizing Industry 5.0 technologies, tools and methods under sustainability principles, following circular integration, is to optimize processes and improve Operational and resource efficiency (ORE) to

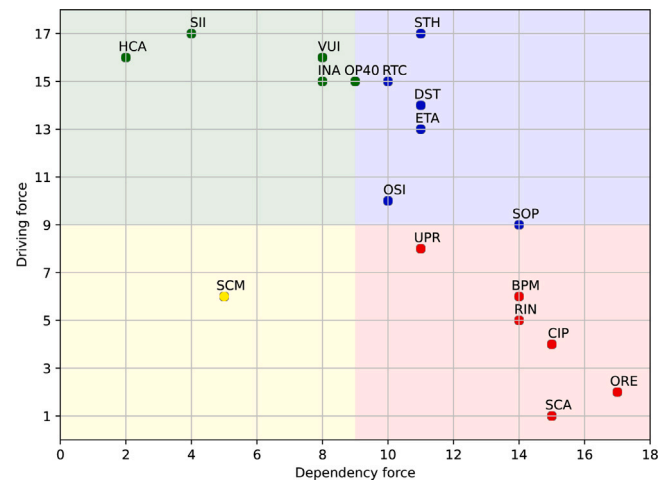


Fig. 5. Classification of Industry 5.0 solutions based on their dependency and driving force.

gain financial, and economic benefits (Commission, 2021b) and improve green total factor productivity (Wang et al., 2023). Furthermore, mass customization initiatives are gaining increasing potential, and organizational Service-orientation and personalization (SOP) abilities are necessary to remain competitive (Durmaz & Kitapci, 2021).

Finally, Supply chain modularity (SCM) refers to the use of flexible and scalable supply chain components that can be easily reconfigured or replaced to accommodate changes in demand or production (Wang & He, 2022). SCM allows for decentralization, and promotes Supply chain adaptability (SCA) to respond to market demands (Wang & He, 2022). Specific enablers include modular automation platforms, digital twins for supply chain design, and adaptive manufacturing systems. The implementation of digital technologies has a positive impact on establishing sustainable supply chain practices (Khan et al., 2023).

The aforementioned Industry 5.0 solutions can be categorized based on their role in organizational transformation, with particular emphasis on their driving and dependency power. Table 1 also reveals the driving and dependency power of Industry 5.0 solutions, which can be categorized based on these two factors. These categories are shown in Fig. 5. This categorization highlights which solutions are pivotal in establishing and stabilizing Industry 5.0 strategies, thereby becoming essential for overall organizational growth and innovation. It also identifies solutions that serve as primary initiators, capable of launching new processes and enhancing existing ones, thus acting as stable drivers of transformation while remaining less susceptible to external fluctuations. Additionally, the categorization reveals solutions that, while not serving as primary catalysts for change, are crucial for supporting and augmenting the capabilities of the more influential solutions. Finally, some solutions are identified as more isolated and specialized, playing a minimal role in shaping the broader strategic direction of Industry 5.0.

The blue cluster includes STH, RTC, DST, ETA, OSI and SOP exhibit strong driving and dependency power, signifying their importance in establishing stable Industry 5.0 strategies and driving transformation. For example, RTC can provide real-time monitoring of organizational processes (BPM), more efficient and automatic data gathering and analysis (INA), monitoring of product life cycle (CIP), etc. (Ghobakhloo et al., 2022). The green cluster includes solutions with strong driving and relatively low dependency force, such as HCA, OP40, VUI, SII, and INA, which play a critical intermediary role in transferring both driving and dependency values but are unstable, meaning that any action taken on them can impact other functions and result in a reflexive effect on themselves. The red cluster includes BPM, CIP, ORE, RIN, SCA, SOP, and UPR exhibit weak driving but strong dependency power, meaning

Table 2
Relationship matrix of sustainability requirements and Industry 5.0 solutions.

GRI topic standard	GRI Topic	Weights	SII	RTC	VUI	DST	INA	OSI	UPR	ETA	SCM	STH	BPM	SOP	RIN	CIP	ORE	SCA	OP40	HCA	
102-16	General disclosure	1		1	1							1									
102-17		1			1							1						1		1	
102-29		1				1						1	1							1	
102-8		1				1							1						1		1
201-1	Economic performance	1	1	1	1	1							1	1	1	1	1	1			
201-2		1											1		1	1					
203-1	Indirect economic impacts	1	1		1	1	1	1		1			1	1	1	1	1				
203-2		1		1	1	1			1		1	1						1		1	
301-1	Materials	1			1							1		1	1	1	1				
301-2		1			1							1		1	1	1	1				
301-3		1			1							1		1	1	1	1				
302-1	Energy	1	1	1	1	1	1					1	1		1	1	1				
302-2		1			1	1		1			1	1			1		1	1			
302-3		1		1	1	1	1						1		1		1				
302-4		1		1	1	1	1						1		1		1				
302-5		1			1	1	1						1		1		1				
306-1	Waste	1				1						1	1		1						
306-2		1	1	1	1	1		1				1	1		1		1				
306-3		1				1						1	1		1						
306-4		1				1						1	1		1						
306-5		1				1						1	1		1						
401-1	Employment	1							1				1				1			1	
401-2		1							1	1			1				1			1	
401-3		1							1				1				1			1	
				4	7	14	15	5	3	4	2	2	14	17	5	16	7	14	3	2	7

that their development relies on various aspects and may require a certain level of maturity to adapt. Finally, the yellow cluster includes SCM, which has weak driving and dependency force.

We must note that the implementation of Industry 5.0 solutions may entail substantial financial investment and necessitate a shift in organizational culture, thereby posing significant challenges (Satyro et al., 2022). To effectively address these challenges, it is essential to prioritize these areas of concern through a coordinated effort. Given the resource-intensive and structurally complex nature of such initiatives, a targeted and strategic approach is essential to effectively manage changes and ensure the achievement of sustainable progress toward desired outcomes.

4.3. Relationship matrix of sustainability requirements and industry 5.0 solutions

The relationship matrix of the application study is shown in Table 2. It identifies the connection between sustainability requirements and Industry 5.0 solutions. This information is stored in an A matrix, where if $a_{i,j} = 1, i = 1, \dots, 24, j = 1, \dots, 18$, the i th requirement can be improved and influenced by the j th solution. The relationships are identified by experts in the field.

The relationship matrix allows us to define which GRI Topic is influenced by which Industry 5.0 solution and also gives us a perspective on how complex the composition of GRI Topics. The last row in Table 2 indicates the importance rank of the Industry 5.0 solutions. The ones with the highest value are the most important solutions to invest in and have a potentially great effect on the selected sustainability requirement. We must note that the interplay between the Industry 5.0 solutions and their effect on each other is not included in the calculation.

The importance ranking of the Industry 5.0 solutions is shown in Fig. 6, which provides an insight into their composition.

In the application study, the most influential Industry 5.0 solution for sustainability requirements is Business process monitoring (BPM), Renewable integration (RIN), and Data sharing, and transparency (DST). However, concerning all the GRI Topics not just the

smaller-scale study, Sustainable thinking (STH) had the most significant influence, followed by DST, BPM, the human-centric management (HCA), then Operational and resource efficiency (ORE), and RIN. Fig. 6 also indicates the composition of the importance rank, including how much and which GRI Topics are influenced by the given Industry 5.0 solution. For example, BPM is the most influential solution, and influences the most GRI Topics/sustainability requirements. The more complex the composition of the Industry 5.0 solution, the more area it can be utilized and might be more general. On the contrary, for example, OP40 is the least influential and only includes the Employment requirements, but it is highly focused and specific.

Regarding a recent study, it has been revealed that 85% of environmental, 65% economic, and 50% societal GRI standards are influenced by Industry 5.0 developments, such as advanced robotics, additive manufacturing, augmented reality, simulation, IoT, big data analytics (related to INA, RTC, ORE), horizontal and vertical integration (related to SII, VUI, SCA), cyber security (related to DST) (Narula et al., 2021). Considering the relationship matrix, more precisely, GRI Topic Standards influenced by the most Industry 5.0 solutions are 203-1 (Infrastructure investments and services supported), 201-1 (Direct economic value generated and distributed), and 302-1 (Energy consumption within the organization). For example, energy consumption within the organization is connected to SII, RTC, VUI, DST, INA, STH, BPM, RIN, CIP, and ORE. The monitoring, control, and optimization of energy consumption have a pivotal role in improving organizational energy efficiency and reducing environmental impact (Ntafalias et al., 2022). There is an increasing need to make buildings and productions cleaner and smarter, manage energy through IoT-based devices, as well as integrate equipment and systems for better operational and resource efficiency (Ntafalias et al., 2022). Energy distribution and management require data acquisition all over the network (Treytl et al., 2004), which demands efficient automated data sharing and real-time communication to support energy management and the use of advanced energy consumption strategies and optimization methods (Maddikunta et al., 2022). Furthermore, smart and connected sensors can provide a real-time overview of the organizational environment, e.g. climate, temperature, energy consumption, etc. (Riedl et al., 2020), which information can support energy management strategies to prevent losses and improve performance and optimize source-productivity (Adel, 2022).

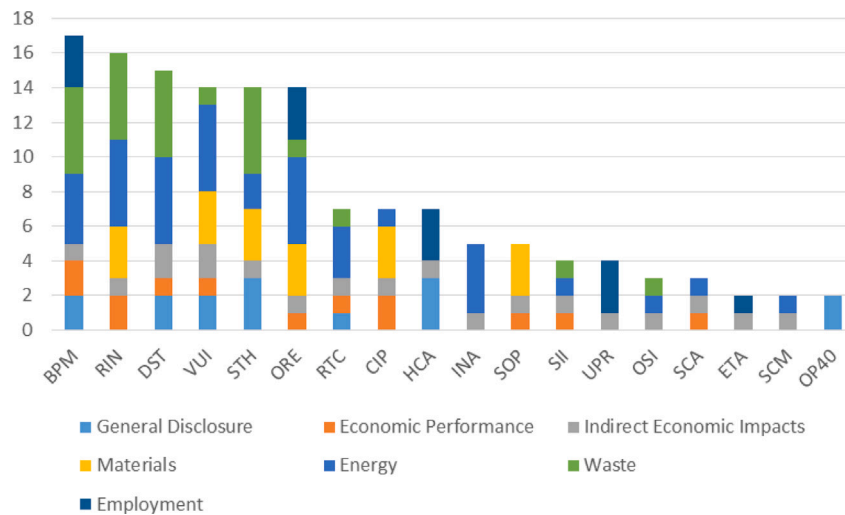


Fig. 6. Importance ranking of Industry 5.0 solutions and their composition.

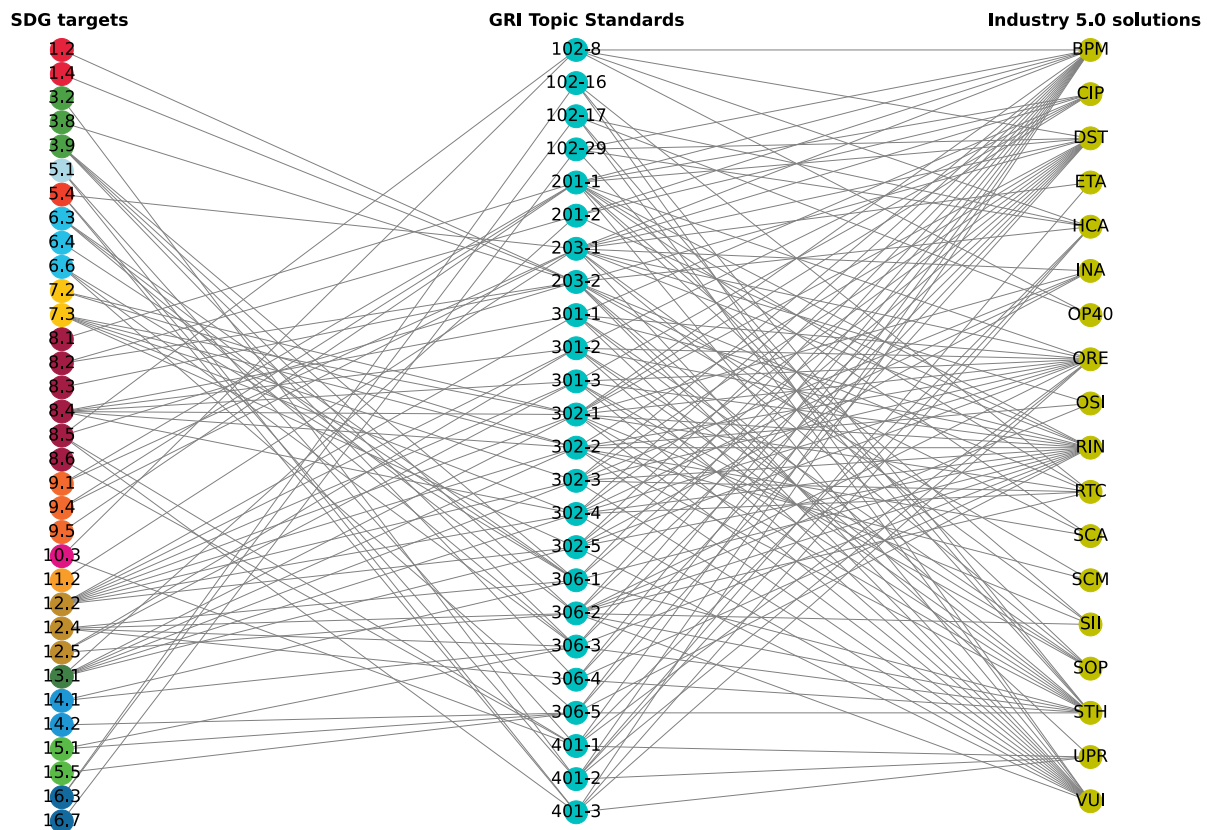


Fig. 7. Bipartite network representation of the connections between SDG targets-GRI Topic Standards -Industry 5.0 solutions.

For a better understanding of the importance of the elements and their structure, the connections of the SDGs-GRI Topics-Industry 5.0 solutions are visualized as bipartite networks in Fig. 7. The edges represent the interaction points between SDGs and GRI Topics, so which topic supports the development of a certain SDG, and the linkages between GRI Topics and the Industry 5.0 solutions, so which development tool can be utilized for improving on a topic.

The network representation can be quite complex in the case of increasing the selected GRI topics. In such cases, a bipartite network projection offers a valuable analytical framework for examining the interlinkages between node sets, such as SDG Targets and GRI Topic Standards, as well as GRI Topic Standards and I5.0 solutions. The

projection of a bipartite network onto one set enables to represent the relationships among the nodes based on their common connections to nodes in the other set. By analyzing the structure of the projected network, we can identify e.g. clusters of closely related topics. This approach facilitates a deeper understanding of the relationships within the dataset, making it possible to draw insights into the multidimensional nature of sustainability reporting practices and the interconnectedness of sustainability issues.

The bipartite representation of the SDG Targets and GRI Topic Standards network (right side of the network represented in Fig. 7) is shown in Fig. 8. In Fig. 8(a) the bipartite network is projected onto the SDG Targets, while in Fig. 8(b) onto the GRI Topics Standards. The node

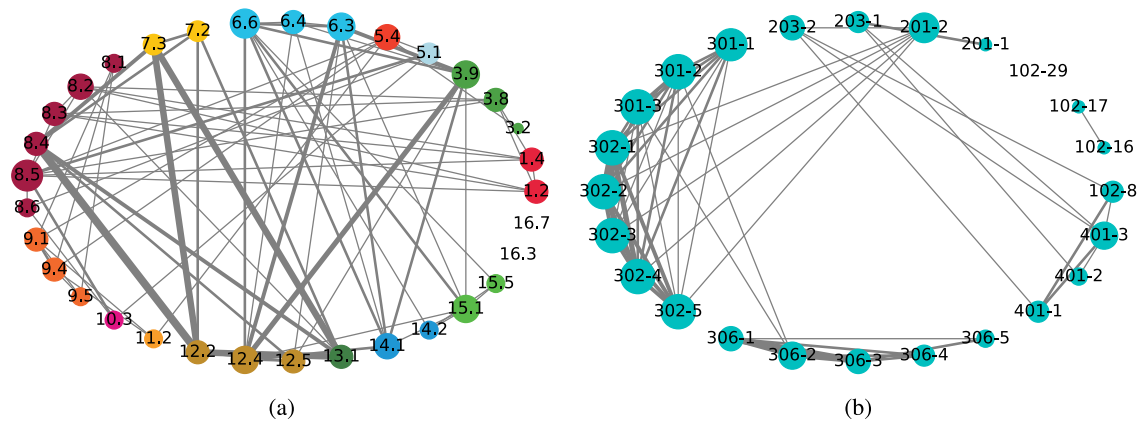


Fig. 8. Projected network of (a) SDG Targets, (b) GRI Topic Standards from the bipartite network of SDGs-GRI Topics positioned in circular layout for better readability. The node sizes represent the centrality of the node, and the thickness of the edges represents the weights, which are the number of linkages between two nodes.

sizes represent the centrality of the node, and the thickness of the edges represents the weights, which are the number of linkages between two nodes, therefore the weight shows how many (a) GRI Topic Standards, (b) SDG Targets two nodes are connected through.

In Fig. 8(a) SDG Targets are linked to each other if they are both linked to the same GRI Topic Standard. This means that the connected SDG Targets are related through a common reporting standard. In this subfigure, the most central element is SDG 8.5 - “Full employment and decent work with equal pay” with a value of 0.28, followed by SDG 6.6 - “Protect and restore water-related ecosystems” with a value of 0.25. Then, SDG 3.9, SDG 6.3, SDG 6.3, SDG 12.4 and SDG 15.1 has centrality 0.21875. Those nodes that are not linked together are connected to GRI Topic Standards that have no influence on other SDG Targets, e.g. SDG 16.7 - “Ensure responsive, inclusive, participatory and representative decision-making at all levels” is linked to GRI 102 - 29 - “Identifying and managing economic, environmental, and social impacts”, but this GRI has no other related SDG Targets in the example.

In Fig. 8(b) GRI Topic Standards are linked to each other if they are both linked to the same SDG Target. This implies that the connected GRI Standards are relevant for reporting on the same SDG Target.

The bipartite representation of the GRI Topic Standards and Industry 5.0 developments network (right side of the network represented in Fig. 7) is shown in Fig. 9. In Fig. 9(a) the bipartite network is projected onto the GRI Topic Standards, while in Fig. 9(b) onto the Industry 5.0 solutions. The node sizes represent the centrality of the node, and the thickness of the edges represents the weights, which are the number of linkages between two nodes, therefore the weight shows how many (a) Industry 5.0 solutions, (b) GRI Topic Standards two nodes are connected through. Both Figures are highly interlined showing the complexity of the network.

In Fig. 9(a) GRI Topic Standards are linked to each other if they are both linked to the same Industry 5.0 solution. This means that the connected GRI Topic Standards are related through a common Industry 5.0 solution. The nodes are highly interlinked since the Industry 5.0 solutions contribute to various GRI sustainability requirements. One of the most central elements is energy (GRI 302) and waste (GRI 306)-related GRI Topic Standards. One of the most linked nodes are GRI 302-1 “Energy consumption within the organization” and GRI 306-2 “Management of significant waste-related impacts” through eight Industry 5.0 solutions.

In Fig. 9(b) Industry 5.0 solutions are linked to each other if they are both linked to the same GRI Topic Standard. This means that the connected SDG Targets are related through a common reporting standard. Value network integration (VUI), and Data sharing and transparency (DST) are the most central elements with value 1.0, followed by Business process monitoring (BPM), Operational and resource efficiency (ORE) and Sustainable thinking (STH) with the value of 0.9412.

BPM and DST has the most linkages between them with the value of 13, meaning that they connect through 13 GRI Topics Standards. It is followed by DST and RIN with the value of 12, ORE and RIN, ORE and VUI, and RIN and VUI with the value of 11.

Communities within the two bipartite network is detected as shown in Fig. 10. The communities in the SDG Targets and GRI Topic Standards network is in Fig. 10(a), while the communities in the GRI Topic Standards and Industry 5.0 solutions network is shown in Fig. 10(b). The colored labels (A-G) represent distinct communities, with each color corresponding to a specific cluster of interconnected elements within the bipartite networks. It is important to note that while the same color scale is used in both subfigures, the colors do not imply that the same communities exist across both networks. Instead, the colors are used to identify different clusters within each network independently. The grayscale shading in the matrices further emphasizes areas of interaction within these networks.

The communities emphasize certain sustainability issue-related developments, such as in Fig. 10(a), the orange ‘A’ community emphasizes enhancing infrastructure, supporting vulnerable populations, and fostering innovation and productivity in the economy. The light green ‘B’ community highlights the importance of integrating sustainability and social responsibility into organizational reporting practices through reporting on workforce composition, employment patterns, and equality measures and ensuring gender equality, decent work, economic growth, and reduced inequalities. The green ‘C’ community emphasizes the importance of environmental sustainability, waste management, and the protection of ecosystems. The associated GRI standards (306 series) highlight a structured approach to reporting on waste management impacts, reinforcing the goals of reducing pollution and enhancing ecosystem resilience. These efforts are crucial for ensuring a healthy environment and a sustainable future. The greenish blue ‘D’ community focuses on improving health outcomes, valuing unpaid care work, fostering economic growth, developing sustainable and resilient infrastructure, and enhancing scientific research and technological innovation. The associated GRI standards emphasize the economic impact and benefits of the initiatives, highlighting how sustainable practices in infrastructure and labor can support broader socio-economic goals. Furthermore, organizations can track their contributions to public health, gender equality, sustainable economic growth, and resilient infrastructure development through these measures. The blue ‘E’ community focuses on goals related to strong institutions, justice, and ethical business practices. These measures are essential for fostering a transparent, just, and accountable environment. The light purple ‘F’ community fosters transparent, inclusive, and accountable decision-making structures, which are vital for sustainable development and the promotion of justice, equity, and participation at all levels of governance. The purple ‘G’ community emphasizes SDG targets that focus on enhancing

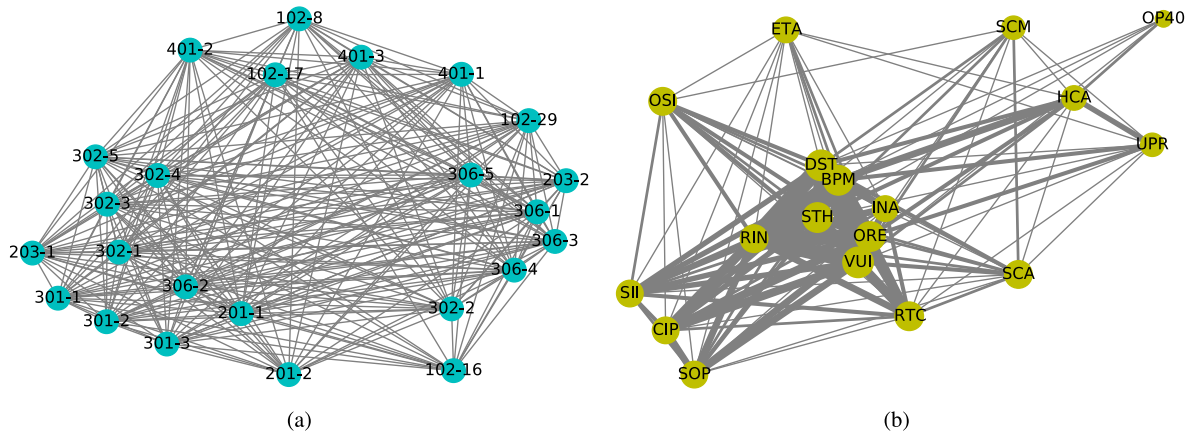


Fig. 9. Projected network of (a) GRI Topic Standards, (b) Industry 5.0 solutions from the bipartite network of GRI Topics-Industry 5.0 solutions. The node sizes represent the centrality of the node, and the thickness of the edges represents the weights, which are the number of linkages between two nodes.

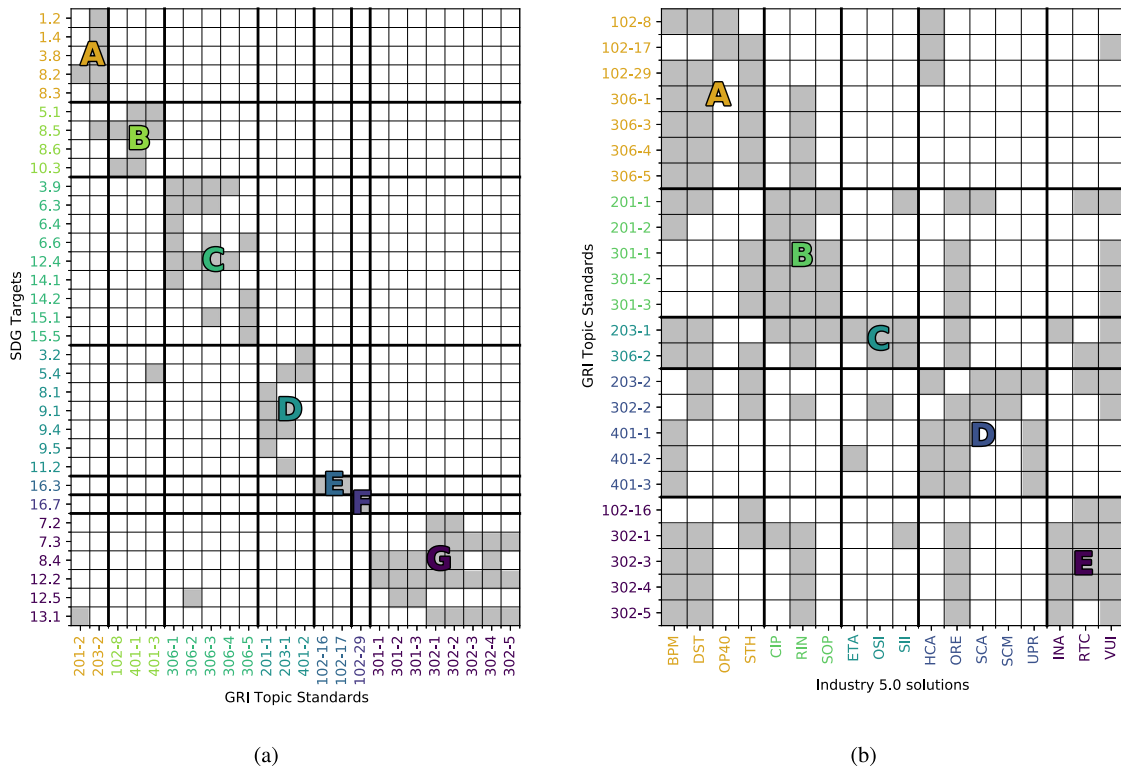


Fig. 10. Communities in the bipartite networks (a) SDG Targets and GRI Topic Standards, (b) GRI Topic Standards and I5.0 solutions. Items in the same communities are indicated with matching colors.

renewable energy use, energy efficiency, sustainable resource management, waste reduction, and climate resilience. The associated GRI standards support these goals by providing metrics for tracking material use, energy consumption, and waste management practices. By aligning organizational practices with these SDG targets and GRI standards, companies can contribute to global sustainability efforts, promote economic growth that is decoupled from environmental degradation, and strengthen resilience against climate-related challenges.

The communities formed in Fig. 10(b) highlight the approach of managing sustainability and operational efficiency, fostering ethical practices, and driving technological advancements. The orange 'A' community focuses on managing ethical, sustainable, and waste-related practices by ensuring efficient and transparent operation and long-term strategic focus on environmental and social impacts and risks. The light green 'B' community focuses on sustainability and innovation

by emphasizing circular intelligent products and renewable energy integration. They prioritize service orientation and personalization to enhance user experience and satisfaction. The community strives to advance environmental responsibility while improving service quality through advanced, tailored solutions. The greenish blue 'C' community focuses on infrastructure development, waste management, and innovation and enhances sustainability through collaborative efforts and efficient use of resources. The blue 'D' community focuses on human-centric management, energy efficiency, and adaptable supply chain practices. It fosters that social well-being and organizational agility can be improved through operational and resource efficiency, enhancing supply chain flexibility, and upskilling the workforce. The purple 'E' community emphasizes ethical standards, energy management, and automation with a focus on reducing energy consumption and aligning operations with sustainable practices.

Table A.1Linked selected sustainability requirements/GRI Topics with SDG indicators and goals based on Ref. [Global Reporting Initiatives \(2022\)](#).

SDG	SDG Targets	Selected GRI topic standards	Description	GRI Topic	
16	16.3	102-16	Values, principles, standards, and norms of behavior	General disclosure	
16	16.3	102-17	Mechanisms for advice and concerns about ethics		
16	16.7	102-29	Identifying and managing economic, environmental, and social impacts		
8	8.5	102-8	Information on employees and other workers		
10	10.3	102-8			
8	8.1	201-1	Direct economic value generated and distributed	Economic performance	
9	9.1	201-1			
9	9.4	201-1			
9	9.5	201-1			
8	8.2	201-2	Financial implications and other risks and opportunities due to climate change		
13	13.1	201-2			
5	5.4	203-1			
9	9.1	203-1	Infrastructure investments and services supported	Indirect economic impacts	
9	9.4	203-1			
11	11.2	203-1			
1	1.2	203-2	Infrastructure investments and services supported		
1	1.4	203-2			
3	3.8	203-2			
8	8.2	203-2			
8	8.3	203-2			
8	8.5	203-2			
8	8.4	301-1	Materials used by weight or volume		Materials
12	12.2	301-1			
8	8.4	301-2	Recycled input materials used		
12	12.2	301-2			
12	12.5	301-2			
8	8.4	301-3	Reclaimed products and their packaging materials		
12	12.2	301-3			
12	12.5	301-3			
7	7.2	302-1	Energy consumption within the organization	Energy	
7	7.3	302-1			
8	8.4	302-1			
12	12.2	302-1			
13	13.1	302-1			
7	7.2	302-2	Energy consumption outside of the organization		
7	7.3	302-2			
8	8.4	302-2			
12	12.2	302-2			
13	13.1	302-2			
7	7.3	302-3	Energy intensity		
12	12.2	302-3			
13	13.1	302-3			
7	7.3	302-4	Reduction of energy consumption		
8	8.4	302-4			
12	12.2	302-4			
13	13.1	302-4			
7	7.3	302-5	Reductions in energy requirements of products and services		
12	12.2	302-5			
13	13.1	302-5			
3	3.9	306-1	Waste generation and significant waste-related impacts	Waste	
6	6.3	306-1			
6	6.4	306-1			
6	6.6	306-1			
12	12.4	306-1			
14	14.1	306-1			
3	3.9	306-2	Management of significant waste-related impacts		
6	6.3	306-2			
12	12.4	306-2			
12	12.5	306-2			
3	3.9	306-3	Waste generated		
6	6.3	306-3			
6	6.6	306-3			
12	12.4	306-3			
14	14.1	306-3			
15	15.1	306-3			

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Table A.1 (continued).

3	3.9	306-4	Waste diverted from disposal	Employment
12	12.4	306-4		
6	6.6	306-5	Waste directed to disposal	
14	14.2	306-5		
15	15.1	306-5		
15	15.5	306-5		
5	5.1	401-1	New employee hires and employee turnover	
8	8.5	401-1		
8	8.6	401-1		
10	10.3	401-1		
3	3.2	401-2	Benefits provided to full-time employees that are not provided to temporary or part-time employees	
5	5.4	401-2		
5	5.1	401-3	Parental leave	
5	5.4	401-3		
8	8.5	401-3		

Table B.1

Indicators for measuring organizational performance on sustainability requirements based on Ref. [Global Reporting Initiatives \(2022\)](#).

GRI topic standards	GRI topic	Description	Indicators from GRI
102-16	General disclosures	Values, principles, standards, and norms of behavior	A description of the organization's values, principles, standards, and norms of behavior.
102-17	General disclosures	Mechanisms for advice and concerns about ethics	A description of internal and external mechanisms for: i. Seeking advice about ethical and lawful behavior, and organizational integrity; ii. Reporting concerns about unethical or unlawful behavior, and organizational integrity
102-29	General disclosures	Identifying and managing economic, environmental, and social impacts	Highest governance body's role in identifying and managing economic, environmental, and social topics and their impacts, risks, and opportunities — including its role in the tabbr implementation of due diligence processes Whether stakeholder consultation is used to support the highest governance body's identification and management of economic, environmental, and social topics and their impacts, risks, and opportunities.
102-8	General disclosures	Information on employees and other workers	Total number of employees by employment contract (permanent and temporary), by gender Total number of employees by employment type (full-time and part-time), by gender Whether a significant portion of the organization's activities are performed by workers who are not employees. If applicable, a description of the nature and scale of work performed by workers who are not employees.
201-1	Economic performance	Direct economic value generated and distributed	a. Direct economic value generated and distributed (EVG&D) on an accruals basis, including the basic components for the organization's global operations as listed below. If data are presented on a cash basis, report the justification for this decision in addition to reporting the following basic components: i. Direct economic value generated: revenues; ii. Economic value distributed: operating costs, employee wages and benefits, payments to providers of capital, payments to government by country, and community investments; iii. Economic value retained: 'direct economic value generated' less 'economic value distributed'. b. Where significant, report EVG&D separately at country, regional, or market levels, and the criteria used for defining significance
201-2	Economic performance	Financial implications and other risks and opportunities due to climate change	a. Direct economic value generated and distributed (EVG&D) on an accruals basis, including the basic components for the organization's global operations as listed below. If data are presented on a cash basis, report the justification for this decision in addition to reporting the following basic components: i. Direct economic value generated: revenues; ii. Economic value distributed: operating costs, employee wages and benefits, payments to providers of capital, payments to government by country, and community investments; iii. Economic value retained: 'direct economic value generated' less 'economic value distributed'. b. Where significant, report EVG&D separately at country, regional, or market levels, and the criteria used for defining significance Risks and opportunities posed by climate change that have the potential to generate substantive changes in operations, revenue, or expenditure, including: i. A description of the risk or opportunity and its classification as either physical, regulatory, or other; ii. A description of the impact associated with the risk or opportunity; iii. The financial implications of the risk or opportunity before action is taken; iv. The methods used to manage the risk or opportunity; v. The costs of actions taken to manage the risk or opportunity

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Table B.1 (continued).

203-1	Indirect Economic Impacts	Infrastructure investments and services supported	a. Extent of development of significant infrastructure investments and services supported. b. Current or expected impacts on communities and local economies, including positive and negative impacts where relevant. c. Whether these investments and services are commercial, in-kind, or pro bono engagements a. Extent of development of significant infrastructure investments and services supported. b. Current or expected impacts on communities and local economies, including positive and negative impacts where relevant. c. Whether these investments and services are commercial, in-kind, or pro bono engagements.
203-2	Indirect economic impacts	Significant indirect economic impacts	Examples of significant identified indirect economic impacts of the organization, including positive and negative impacts Significance of the indirect economic impacts in the context of external benchmarks and stakeholder priorities, such as national and international standards, protocols, and policy agendas
301-1	Materials	Materials used by weight or volume	Total weight or volume of materials that are used to produce and package the organization's primary products and services during the reporting period, by: i. Non-renewable materials used; ii. Renewable materials used
301-2	Materials	Recycled input materials used	Percentage of recycled input materials used to manufacture the organization's primary products and services.
301-3	Materials	Reclaimed products and their packaging materials	Percentage of reclaimed products and their packaging materials for each product category
302-1	Energy	Energy consumption within the organization	Total fuel consumption within the organization from non-renewable sources, in joules or multiples, and including fuel types used In joules, watt-hours or multiples, the total: i. Electricity consumption ii. Heating consumption iii. Cooling consumption iv. Steam consumption In joules, watt-hours or multiples, the total: i. Electricity sold ii. Heating sold iii. Cooling sold iv. Steam sold Total energy consumption within the organization, in joules or multiples. Total fuel consumption within the organization from renewable sources, in joules or multiples, and including fuel types used. Energy consumption outside of the organization, in joules or multiples.
302-2	Energy	Energy consumption outside of the organization	Energy intensity ratio for the organization.
302-3	Energy	Energy intensity	Energy intensity ratio for the organization.
302-4	Energy	Reduction of energy consumption	Amount of reductions in energy consumption achieved as a direct result of conservation and efficiency initiatives, in joules or multiples
302-5	Energy	Reductions in energy requirements of products and services	Reductions in energy requirements of sold products and services achieved during the reporting period, in joules or multiples.
306-1	Waste	Waste generation and significant waste-related impacts	Total volume of planned and unplanned water discharges by: i. Destination; ii. Quality of the water, including treatment method; iii. Whether the water was reused by another organization
306-2	Waste	Management of significant waste-related impacts	Total weight of hazardous waste, with a breakdown by the following disposal methods where applicable: i. Reuse ii. Recycling iii. Composting iv. Recovery, including energy recovery v. Incineration (mass burn) vi. Deep well injection vii. Landfill viii. On-site storage Other (to be specified by the organization)

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5. Discussion and conclusions

The evolution of Industry 5.0 has provided a distinct perspective that highlights the crucial importance of aligning digital transformation with socio-economic and environmental priorities. “Research and innovation drive the transition to a sustainable, human-centric

and resilient” (Commission, 2021b) industrial operations. To achieve such, a shift in consumer behavior, production impact, and governmental policies are required (Küfeoğlu, 2022) and the aligning of sustainability-related metrics and indicators is required to provide a frame for sustainable production processes (Viles et al., 2022).

Table B.1 (continued).

306-3	Waste	Waste generated	Impacts of significant spills
			Total number and total volume of recorded significant spills
306-4	Waste	Waste diverted from disposal	The following additional information for each spill that was reported in the organization's financial statements: i. Location of spill; ii. Volume of spill; Material of spill, categorized by: oil spills (soil or water surfaces), fuel spills (soil or water surfaces), spills of wastes (soil or water surfaces), spills of chemicals (mostly soil or water surfaces), and other (to be specified by the organization)
			Total weight for each of the following: i. Hazardous waste transported ii. Hazardous waste imported iii. Hazardous waste exported Hazardous waste treated
306-5	Waste	Waste directed to disposal	Percentage of hazardous waste shipped internationally
			a. Water bodies and related habitats that are significantly affected by water discharges and/or runoff, including information on: i. The size of the water body and related habitat; ii. Whether the water body and related habitat is designated as a nationally or internationally protected area; iii. The biodiversity value, such as total number of protected species.
401-1	Employment	New employee hires and employee turnover	Total number and rate of employee turnover during the reporting period, by age group, gender and region
			Average hours of training that the organization's employees have undertaken during the reporting period, by: i. Gender; ii. Employee category.
401-2	Employment	Benefits provided to full-time employees that are not provided to temporary or part-time employees	Benefits which are standard for full-time employees of the organization but are not provided to temporary or part-time employees, by significant locations of operation. These include, as a minimum: i. Life insurance; ii. Health care; iii. Disability and invalidity coverage; iv. Parental leave; v. Retirement provision; vi. Stock ownership; vii. Others
			Total number of employees that were entitled to parental leave, by gender
401-3	Employment	Parental leave	Total number of employees that took parental leave, by gender
			Total number of employees that returned to work in the reporting period after parental leave ended, by gender
			Total number of employees that returned to work after parental leave ended that were still employed 12 months after their return to work, by gender
			Return to work and retention rates of employees that took parental leave, by gender
			Percentage of total employees by gender and by employee category who received a regular performance and career development review during the reporting period

To reduce environmental impact, promote worker well-being, and ensure resilient business operations, organizations must establish a culture and strategy that prioritizes sustainability and human-centric principles aligned with Industry 5.0 principles. The systematic development and reporting cycle of Industry 5.0 can assist organizations in assessing their transition progress also in Environmental, Social and Governance practices. However, the contribution of Industry 5.0 toward sustainable development values is currently under-investigated (Ghobakhloo et al., 2022), and decision-support tools are needed to identify which Industry 5.0 solutions can support specific sustainability requirements.

To address this need, we propose a decision support and development tool that is built upon the QFD standardized method.

While previous studies have explored the use of QFD within Industry 5.0, they primarily focus on specific domains such as sustainable supply chain design, often lacking integration with broader organizational performance indicators and reporting frameworks. The novelty of our approach lies in extending the traditional QFD methodology by incorporating standardized sustainability frameworks (GRI standards and SDG targets) and enhancing relational analysis through network science. This method enables a more holistic and systemic identification of how Industry 5.0 solutions align with specific sustainability requirements and organizational goals. Our approach not only identifies key

enablers but also supports decision-making by revealing critical inter-dependencies, driving factors, and performance implications, thereby offering a practical and scalable tool for organizations aiming to align their development with Industry 5.0 and sustainability imperatives. The QFD-based method includes three parts: (1) identification of the connections between SDGs and sustainability requirements based on the GRI Topic Standards (Global Reporting Initiatives, 2022); (2) a causation matrix of Industry 5.0 solutions based on Ghobakhloo et al. (2022), which has been extended with Operator 4.0 and Human-centric approach that identify the driving and dependency factors of each solution and their synergies; (3) a relationship matrix of sustainability requirements and Industry 5.0 that identifies which investments drive which sustainability requirement and organizational performance on SDG goals and targets. The relationship matrix has also been analyzed through network analysis identifying the direct and indirect relationships in the bipartite graphs of SDGs and GRI Topics, and the GRI Topics and Industry 5.0 solutions. Furthermore, identifying communities in the bipartite graphs allows to find closely interlinked sub-networks. The methodology only provides guidance to identify key relationships and potential solutions and tools to develop organizational performance and sustainability and reporting. The methodology

cannot be utilized by itself as a strategic plan, it requires organizational experts and additional considerations for tailored strategy development.

Furthermore, we must note that since in Hungary Industry 5.0 and GRI-based reporting are not yet widely adopted, therefore a complete application study cannot be represented, but an experience-based example. Based on the analysis of the expert experience-based example study to explore key relationships, it was found that the SDGs with the highest level of connectivity to sustainability requirements are SDG8, SDG12, SDG16, and SDG3. However the most central SDG through the sustainability requirements are SDG3 and SDG8, while the least interlinked is SDG16. These findings are aligned with the principles of Industry 5.0, which emphasizes the need to establish strong, resilient institutions that prioritize worker well-being and promote economic growth and productivity through the integration of digital technologies. Additionally, Industry 5.0 promotes sustainable production and aims to decrease environmental impact in areas such as energy, waste, and water management. However, the focus on SDGs and sustainability requirements can differ for each industry and even country due to the industrial and country-specific requirements and development initiatives.

The most critical Industry 5.0 solutions were identified as Sustainable thinking, followed by Data sharing and transparency, Business process monitoring, and Human-centric management, then Operational and resource efficiency, and Renewable integration. These are also the most central elements considering the linkages through sustainability requirements. Furthermore, Sustainable thinking, Data sharing and transparency, System integration, and interoperability have great driving power and low dependence power which indicates that these are foundations of establishing Industry 5.0 strategies and supports other solutions to be efficient. Therefore, investing in these solutions is fundamental.

According to a recent empirical study, which was performed 15 companies, they also revealed that key enablers of sustainable Industry 5.0 transition by categories are [Caiado et al. \(2024\)](#): (1) sustainable corporate culture, communication, management commitment, collaborative networks and strategic alignment which is covering the HCA and STH Industry 5.0 solution; (2) renewable resources, data transparency and flow, internal innovation and business model development, which is in line with RIN, DST, BOM, CIP; (3) balanced integration that is SII, VUI, INA. Furthermore, according to Yada et al. Industry 5.0 applications can provide better reporting in terms of authenticity, retrospective reporting, customizability, scope of reporting, cost and efficacy ([Yadav et al., 2024](#)). The applications highlighted are in line with our study such as data collection at source, advanced data analytical capabilities (related to RTC), authenticated transactions (related to DST), or enhanced integration across devices and entities (related to SII). Considering these enablers of Industry 5.0 organizational transformation, it aligns with the aspiration of achieving a balanced development of both technological capabilities and human-centric organizational management, and develop solutions focusing on human aspects. In order to emphasize human aspirations, additional two Industry 5.0 solutions, the Operator 4.0 and Human-centric approach have to gain attention. From both aspects, since the former focusing on the technicalities, while the latter on the managerial and strategic aspects of creating an ethical, transparent, learning organizational environment. We must note that each solution can be further decomposed, which we aimed to highlight in the exploration study.

It should be emphasized that investing in digital technologies is essential for organizations to retain competitiveness and satisfy sustainability requirements. Nevertheless, the applicability of such technologies, tools, and methods is reliant upon the specific organizational context and targeted business objectives. Moreover, it is becoming increasingly clear that investing in emerging technologies alone will not suffice; rather, a move toward cross-collaboration will be necessary to attain a shared objective, such as energy-efficient and

cleaner production. Achieving such goals necessitates the establishment of digital trust, security, and an organizational culture that supports collaboration.

We believe that the presented aspect and method support organizational strategy-making that centers on Industry 5.0 principles while identifying sustainability requirements and performances on SDGs. In addition, it enables the selection of Industry 5.0-driven tools and techniques to invest in and integrate into organizational processes to achieve targeted business objectives. Furthermore, allows for the measurement and reporting of organizational performance, and allows benchmarking activities.

CRediT authorship contribution statement

Tímea Czvetkó: Writing – original draft, Visualization, Validation, Software, Methodology, Formal analysis. **Viktor Sebestyén:** Writing – review & editing, Writing – original draft, Validation, Supervision, Methodology, Funding acquisition, Conceptualization. **János Abonyi:** Writing – review & editing, Validation, Supervision, Methodology, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Linked sustainability requirements and SDG goals and indicators

See [Table A.1](#).

Appendix B. Indicators of measuring organizational performance

See [Table B.1](#).

Data availability

All data have been shared within the manuscript.

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